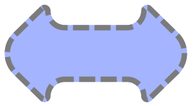
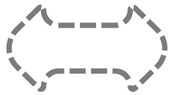
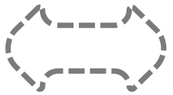
Lightning gate

	Light blue	dark blue
on		
off		

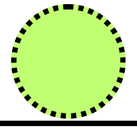
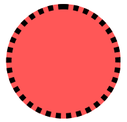
properties	• active: binary on/off states • colour: light blue or dark blue

behaviors	• when on
	  ◦ stops player from walking through it
	  ◦ destroys players and boxes that are in the same space.

game static feedback	• visual
	  ◦ colour changes to a white when off
	• sound
	  ◦ sound effect on state change

systemetic feedback	• stops player and box from going through when on • destroys player / box in area when on. • projectiles can pass through while active

Pressure Button

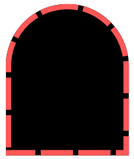
on	
off	

properties	• Toggle state: Binary state; on / off • colour: green of red
behaviors	• Toggles to on state when player or a box is on it.

game static feedback	• visual
	  ◦ colour changes to match state colour
	• sound
	  ◦ click sound when toggled

systemetic feedback	toggle affects wire object.

Exit Door

open	
closed	

properties	• active: binary open/closed states
	• colour: white or black depending on state
	  ◦ matches state of connected wire
	• stroke: green or red depending on state
	  ◦ accessibility, dotted or full stroke depending on state

behaviors	• when open ◦ touching it will bring the player to the next level

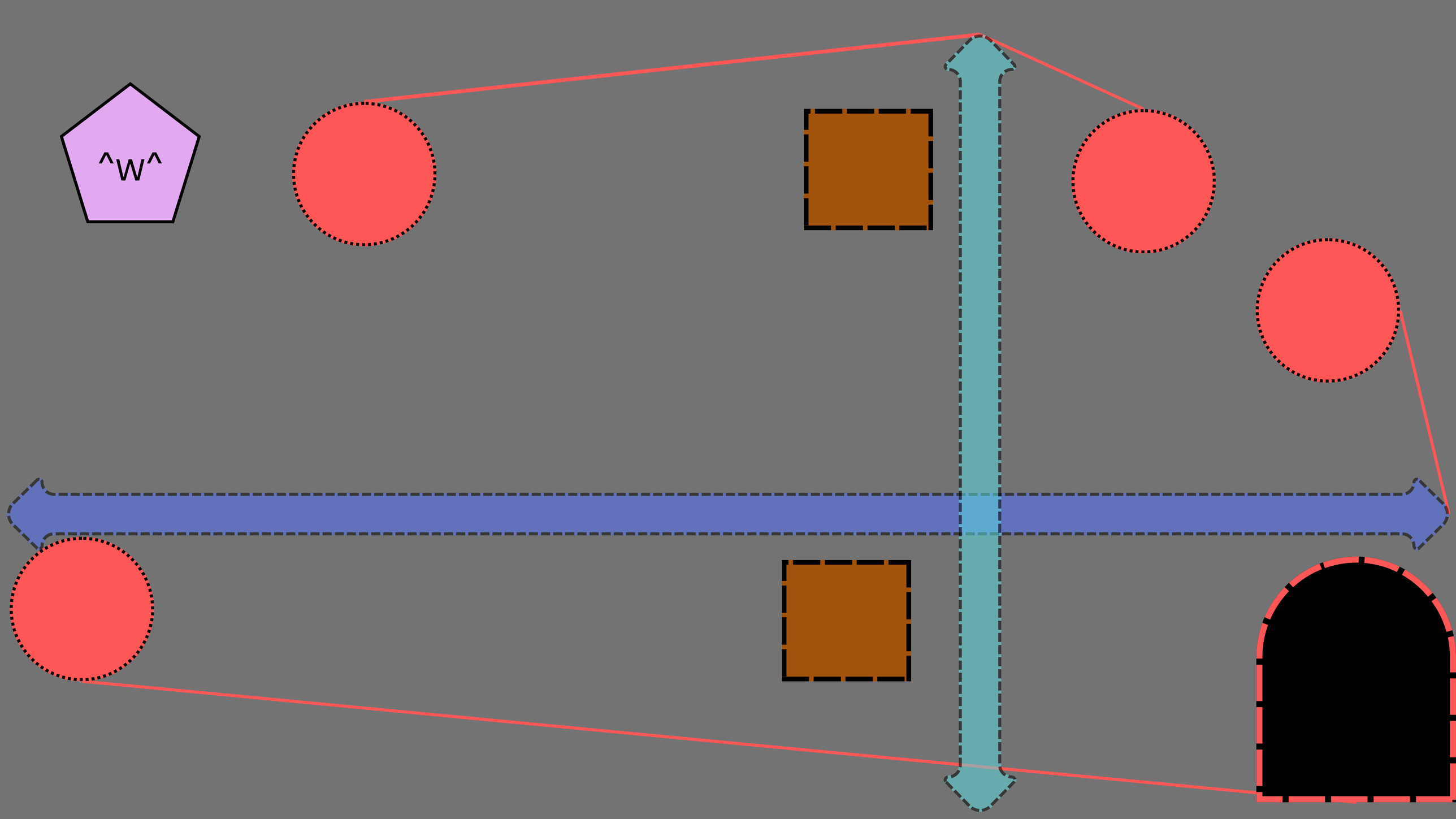
game static feedback	• visual
	  ◦ colour changes to match state colour
	• audio
	  ◦ door unlock sound when state changes

systemetic feedback	• when open, touching it will bring player to next level.

Wire

on	
off	

properties	• Toggle state: Binary state; on / off • colour: green of red
behaviors	• toggles state to match the connected button
game static feedback	• visual ◦ colour changes to match state colour
systemetic feedback	Wire toggles connected objects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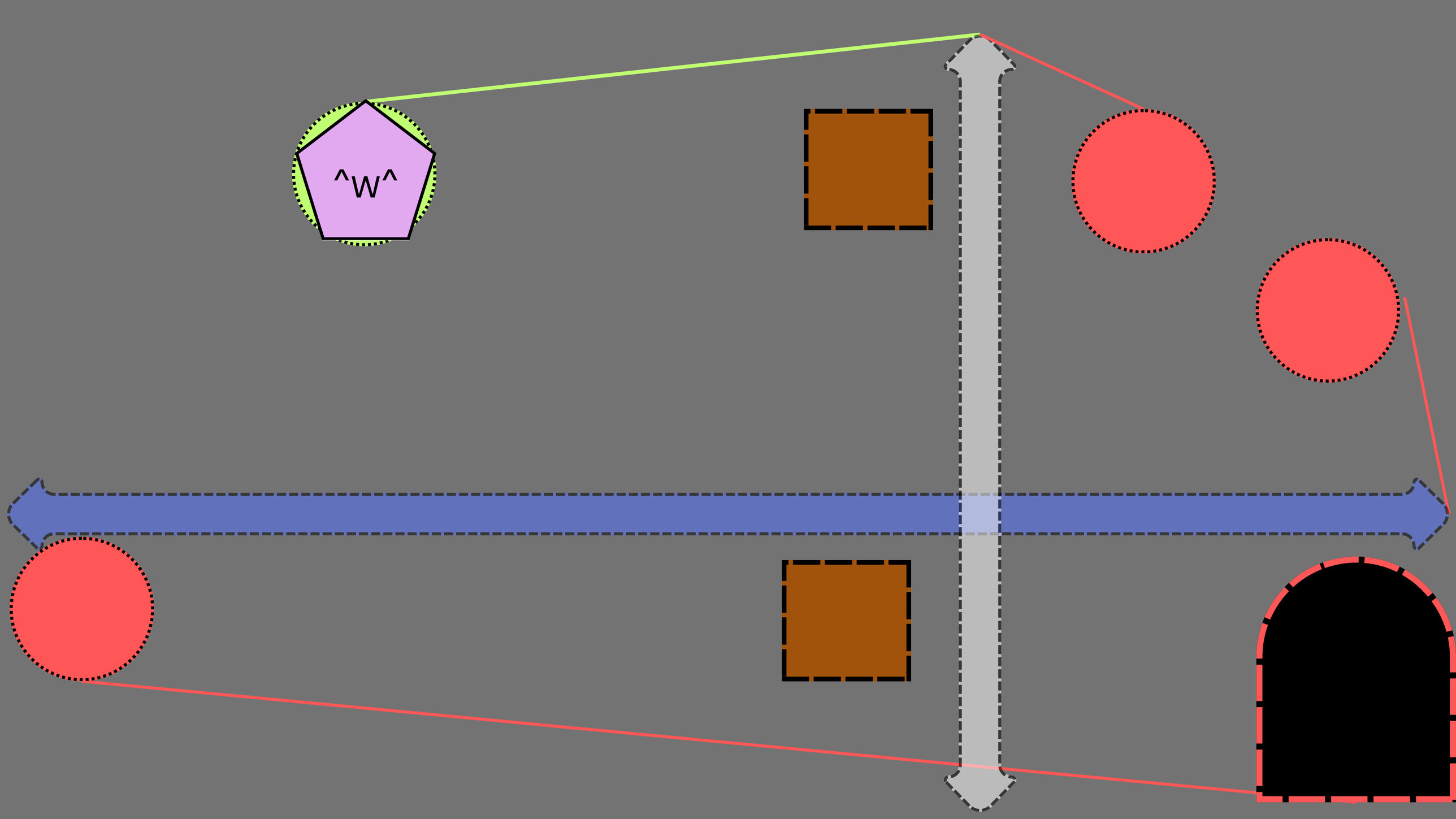
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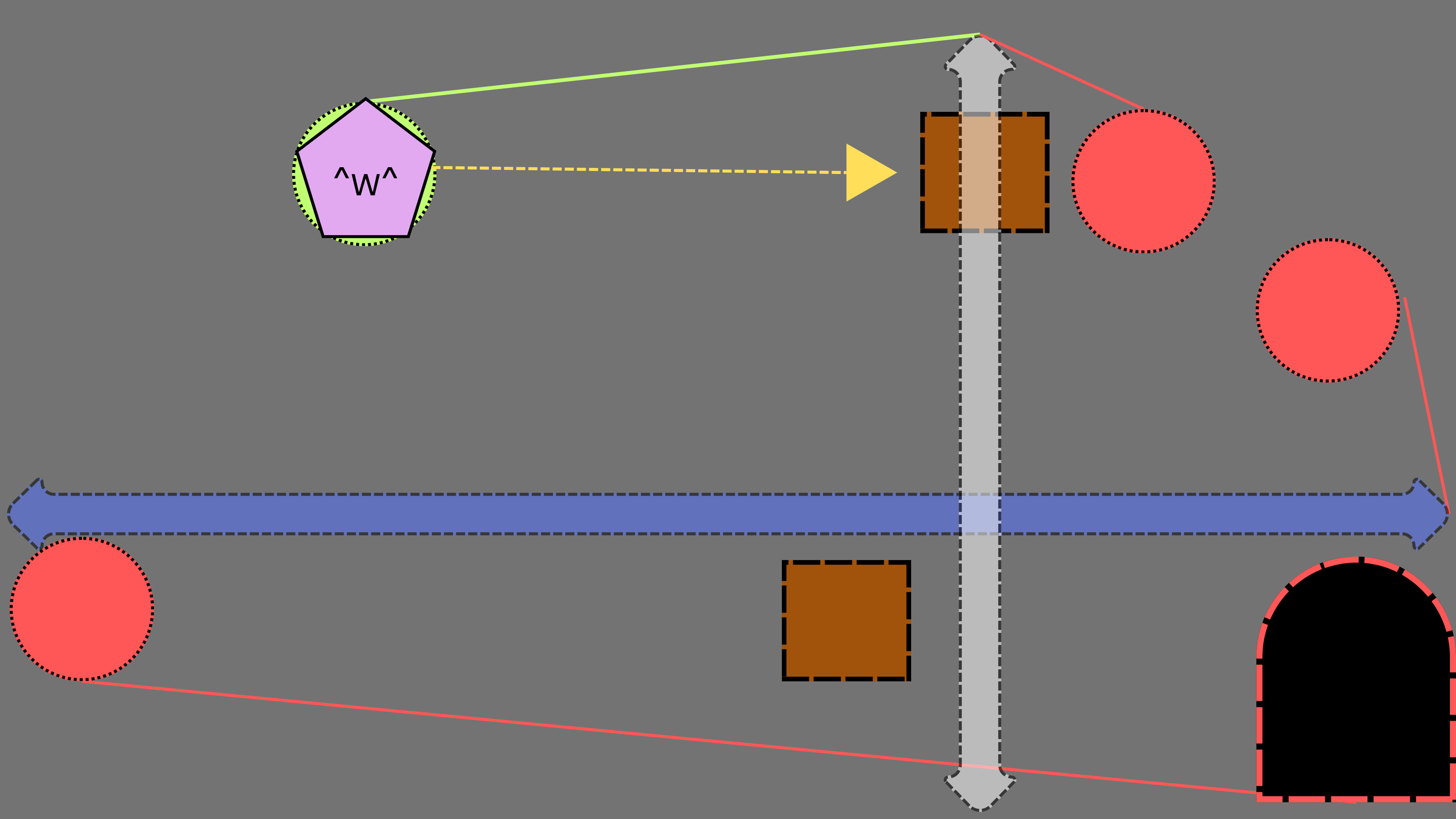
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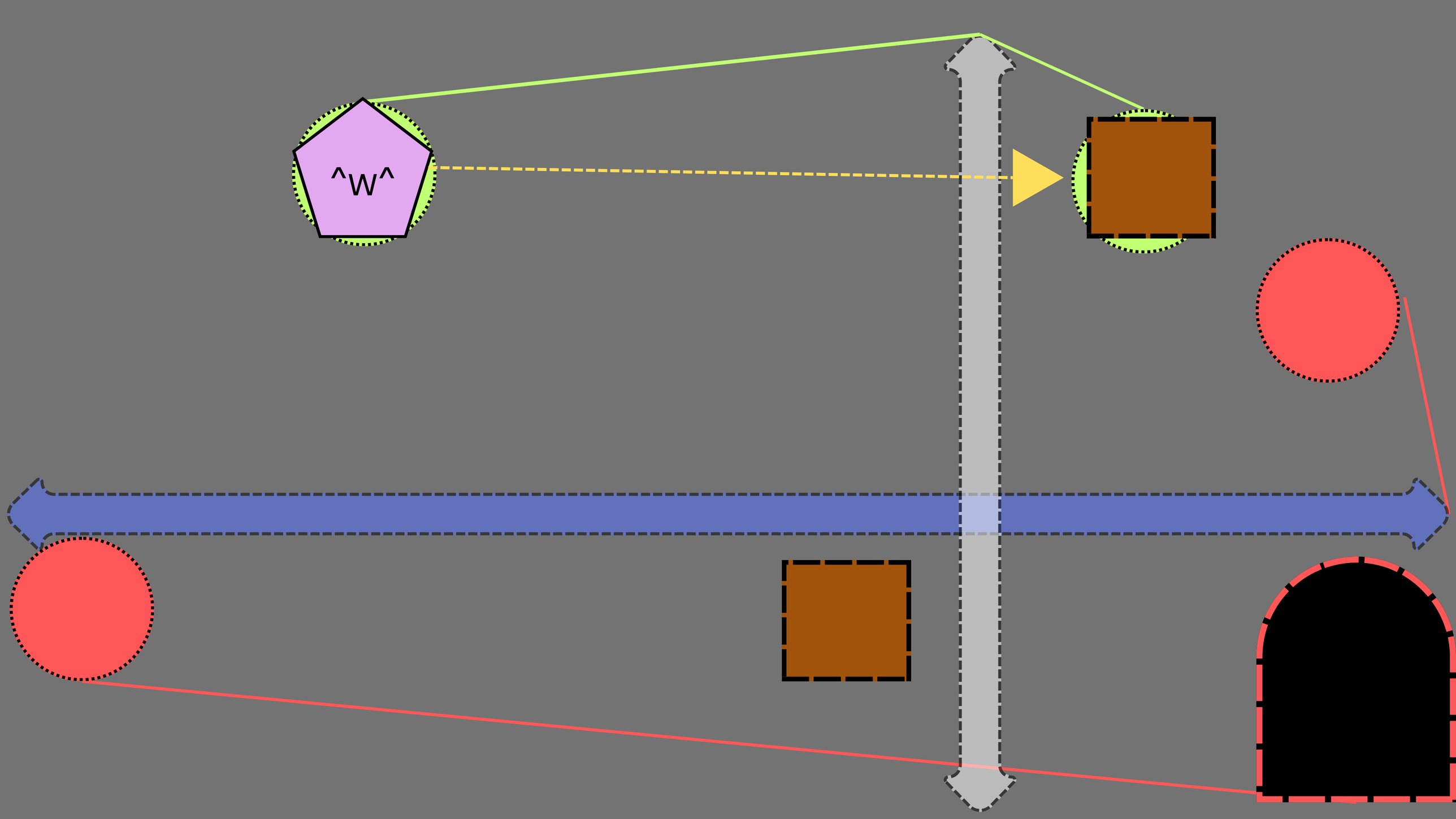
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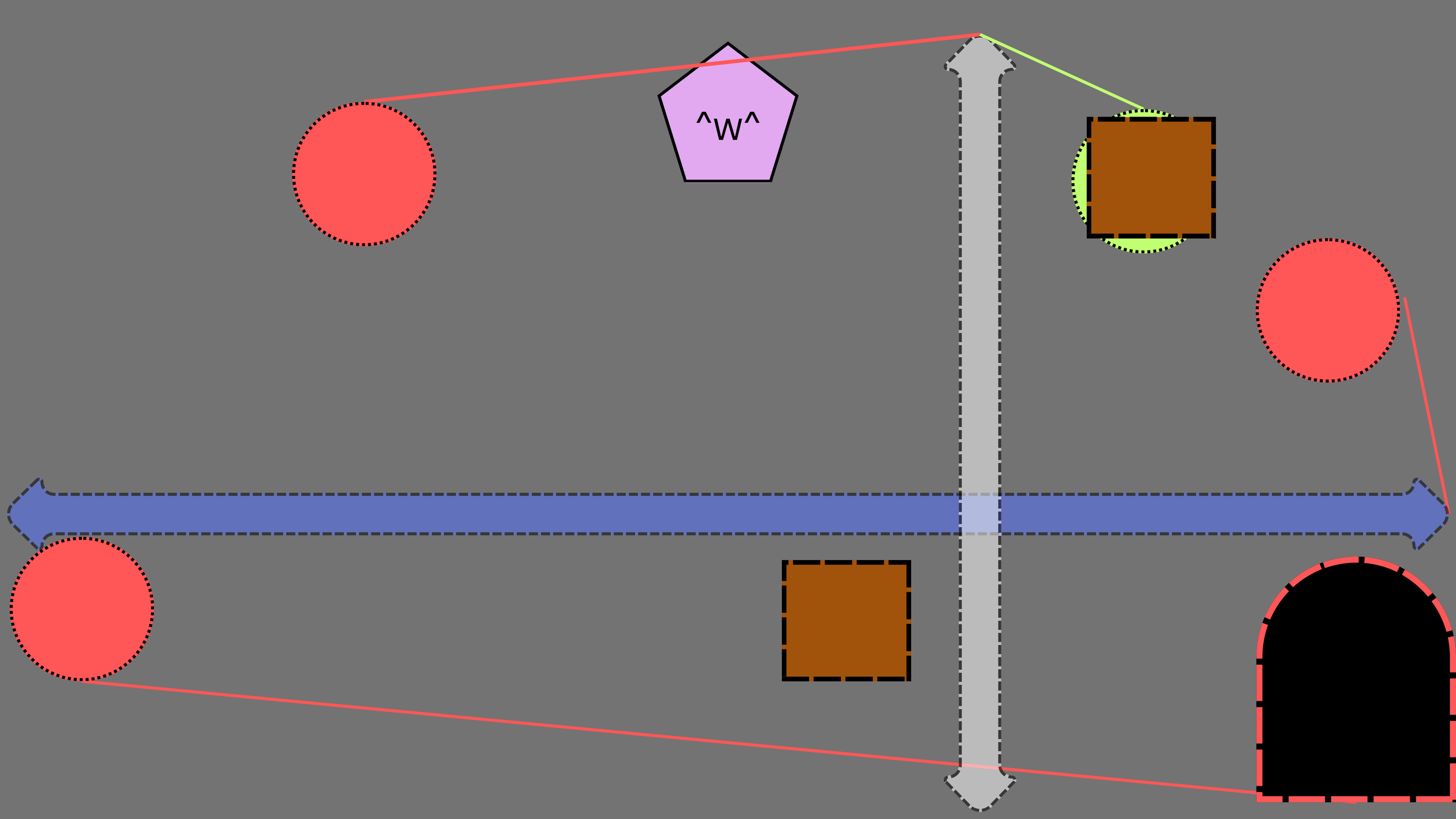
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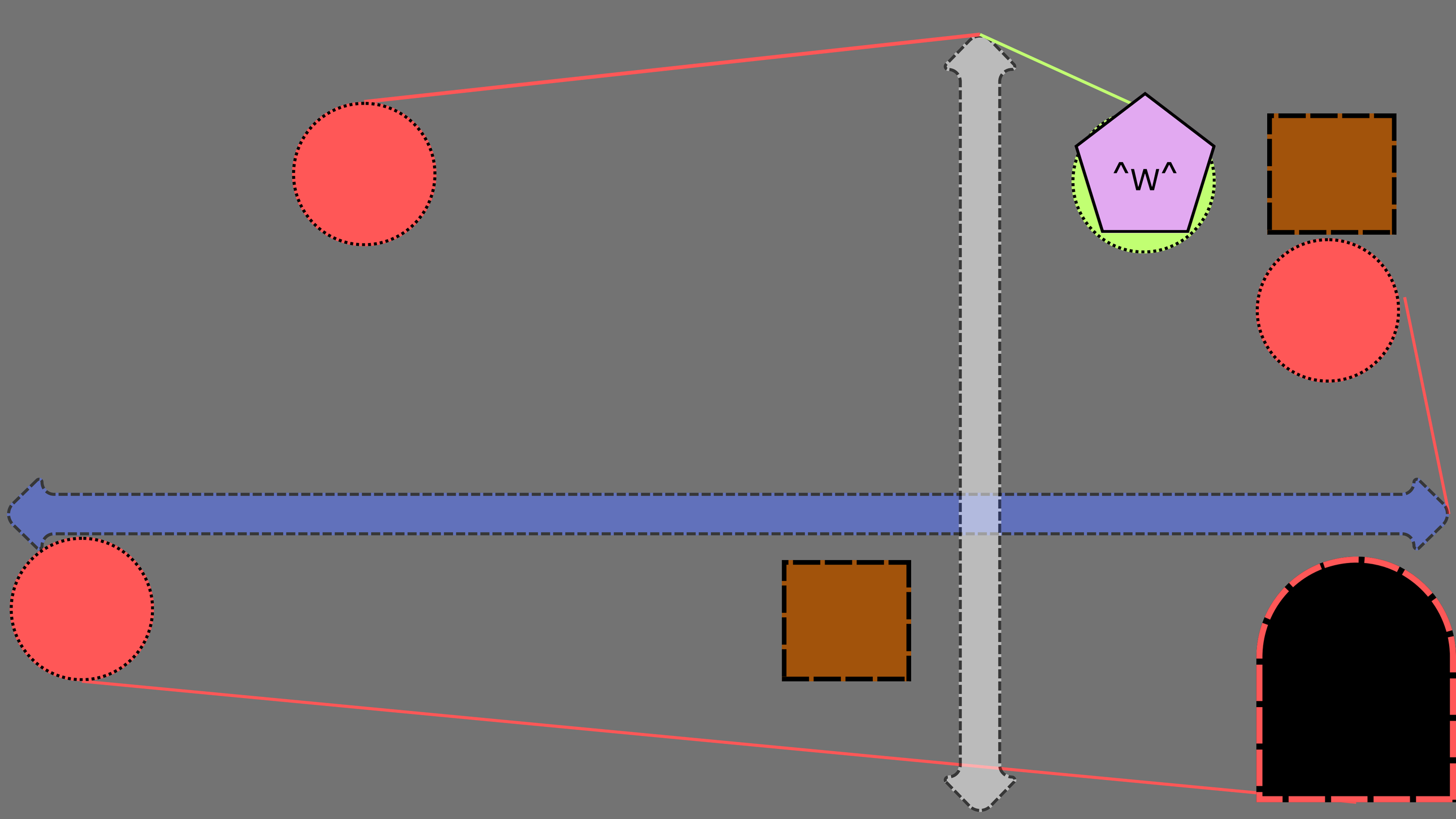
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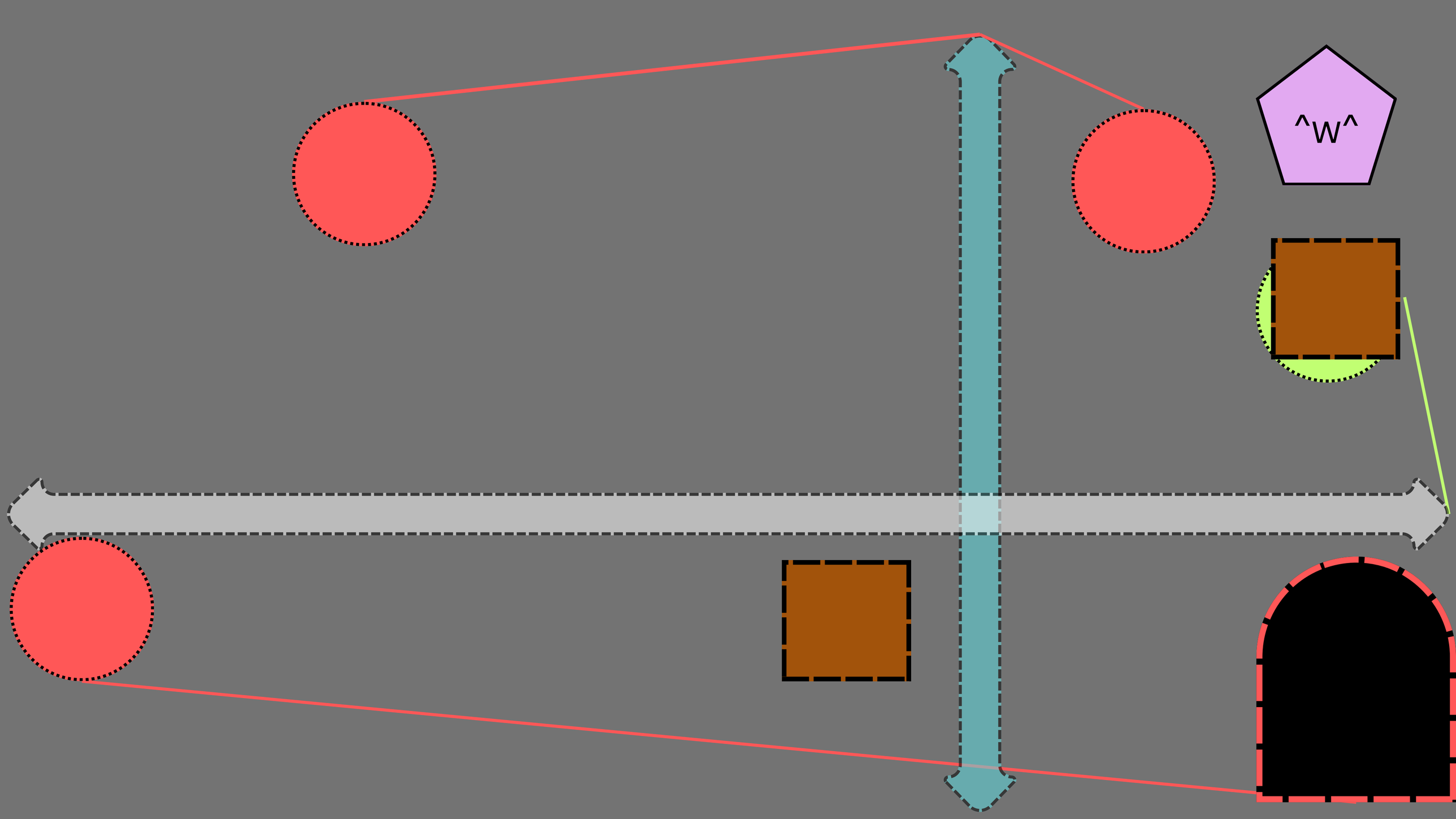
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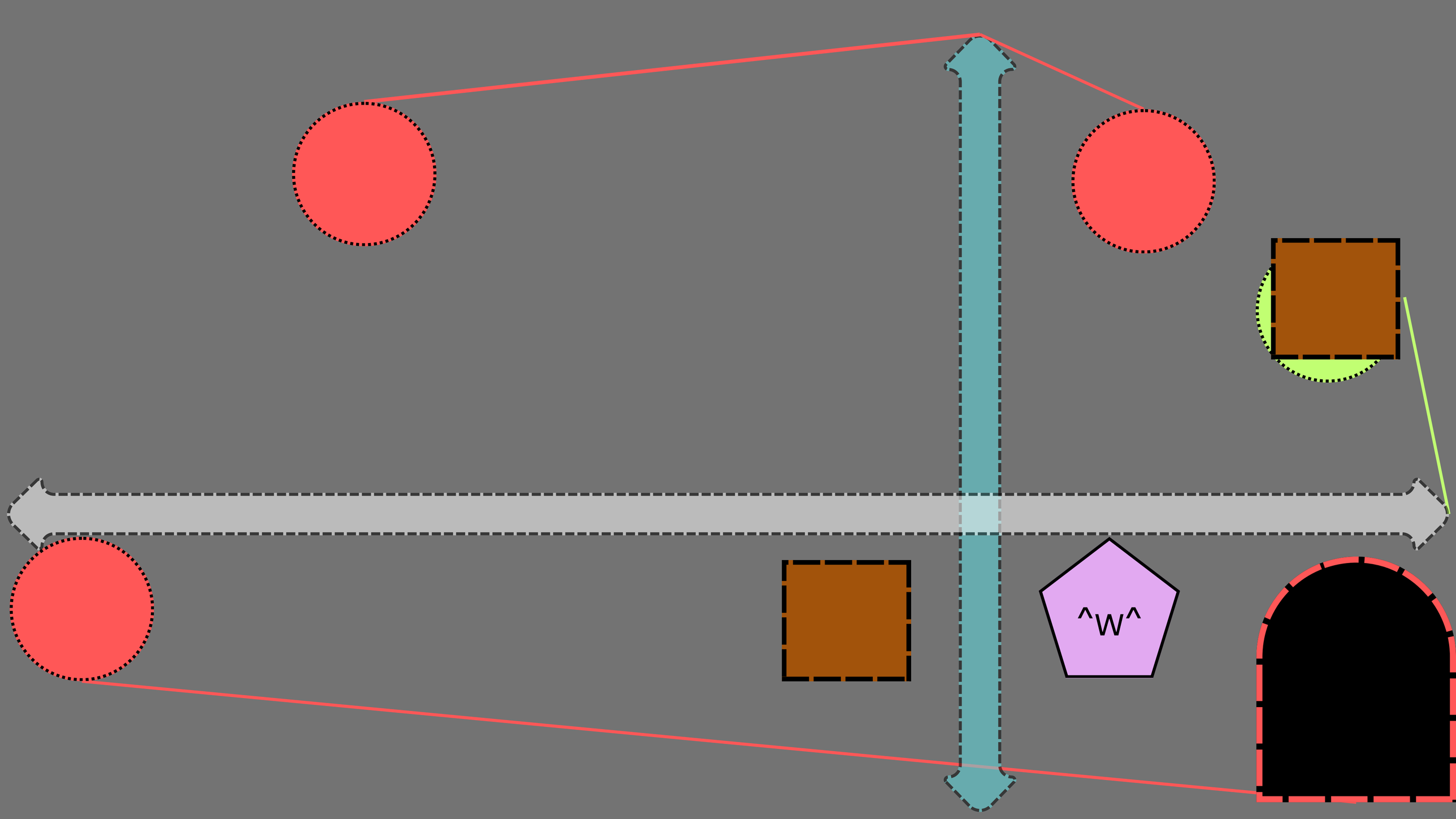
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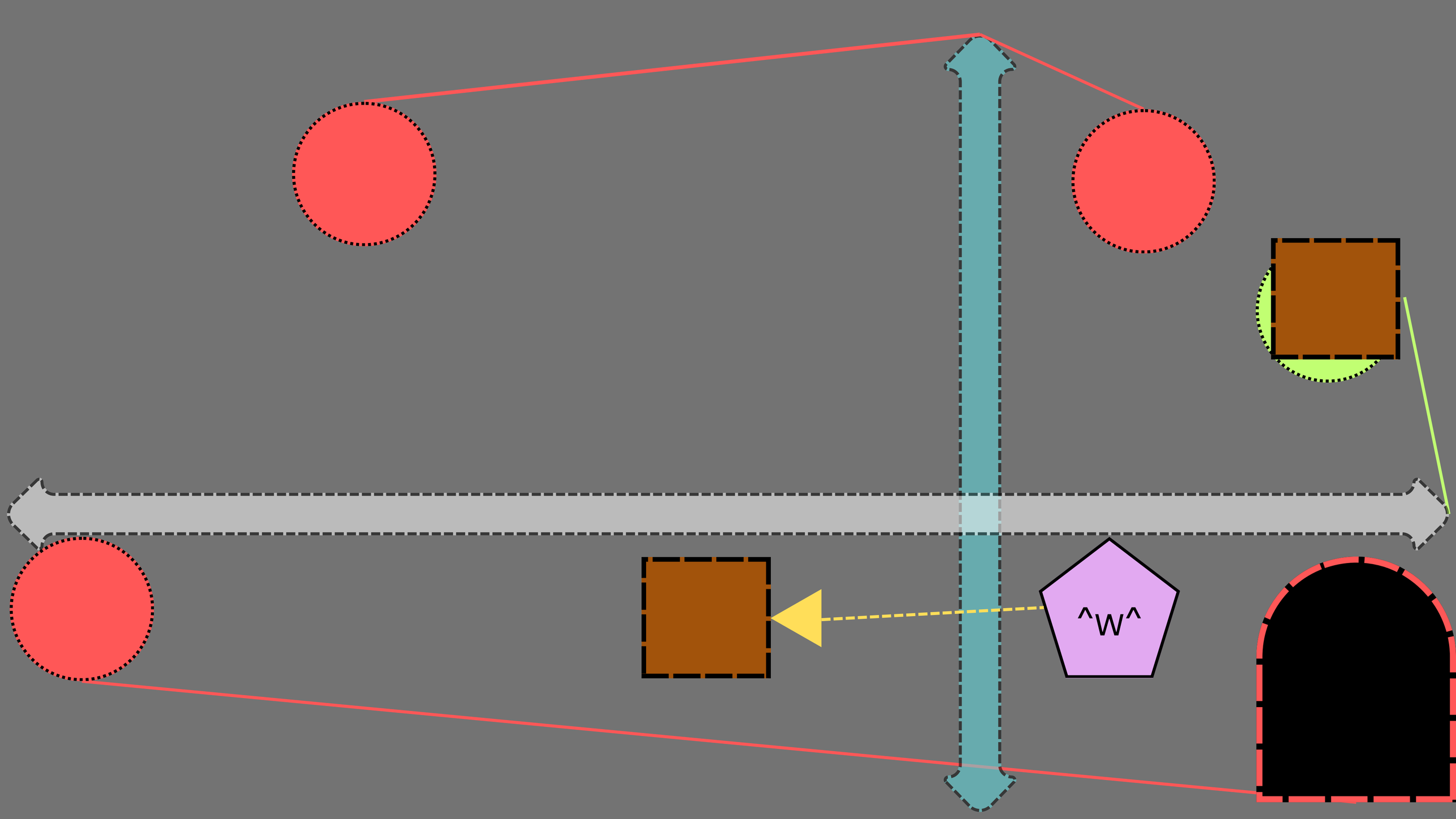
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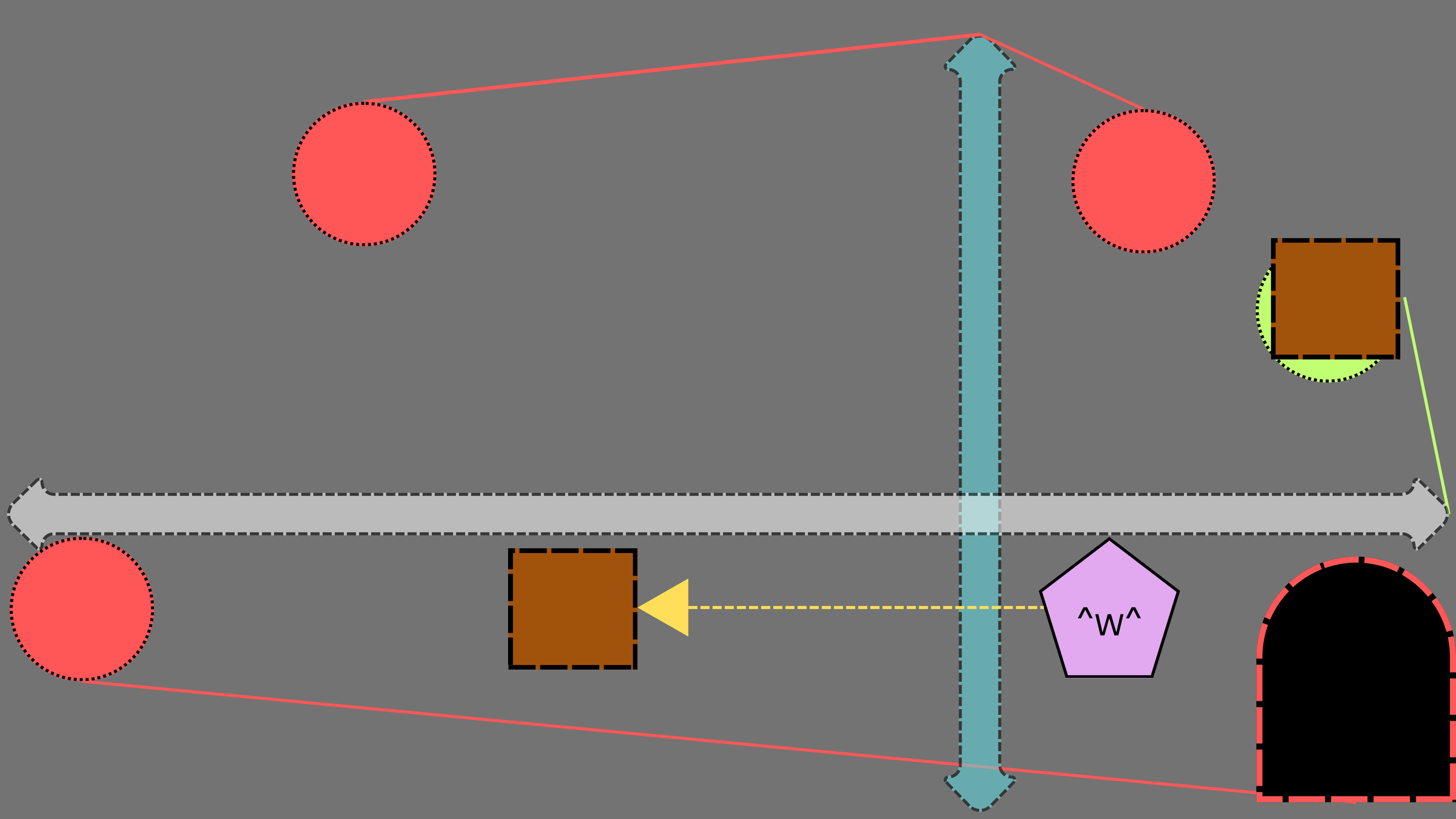
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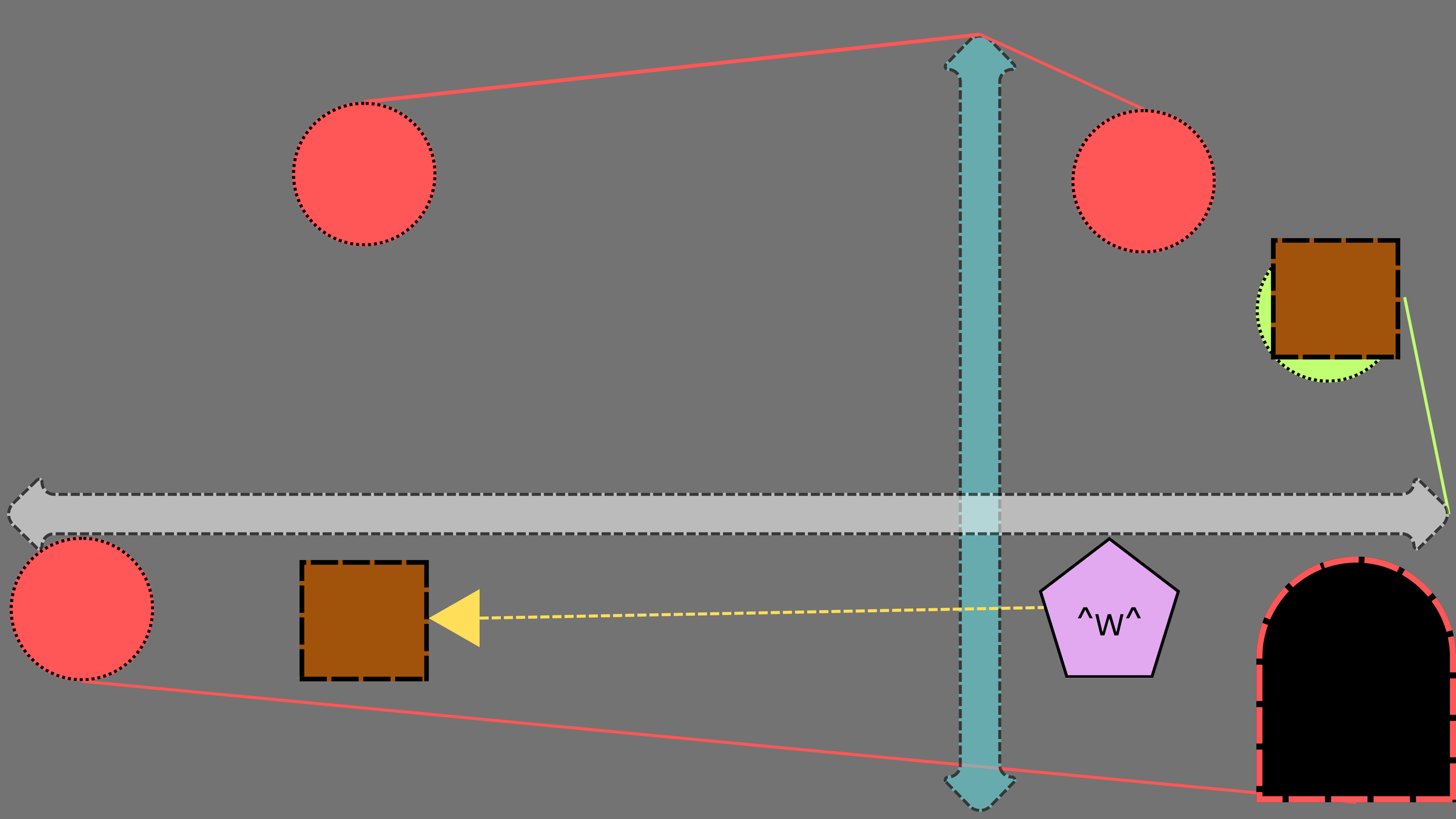
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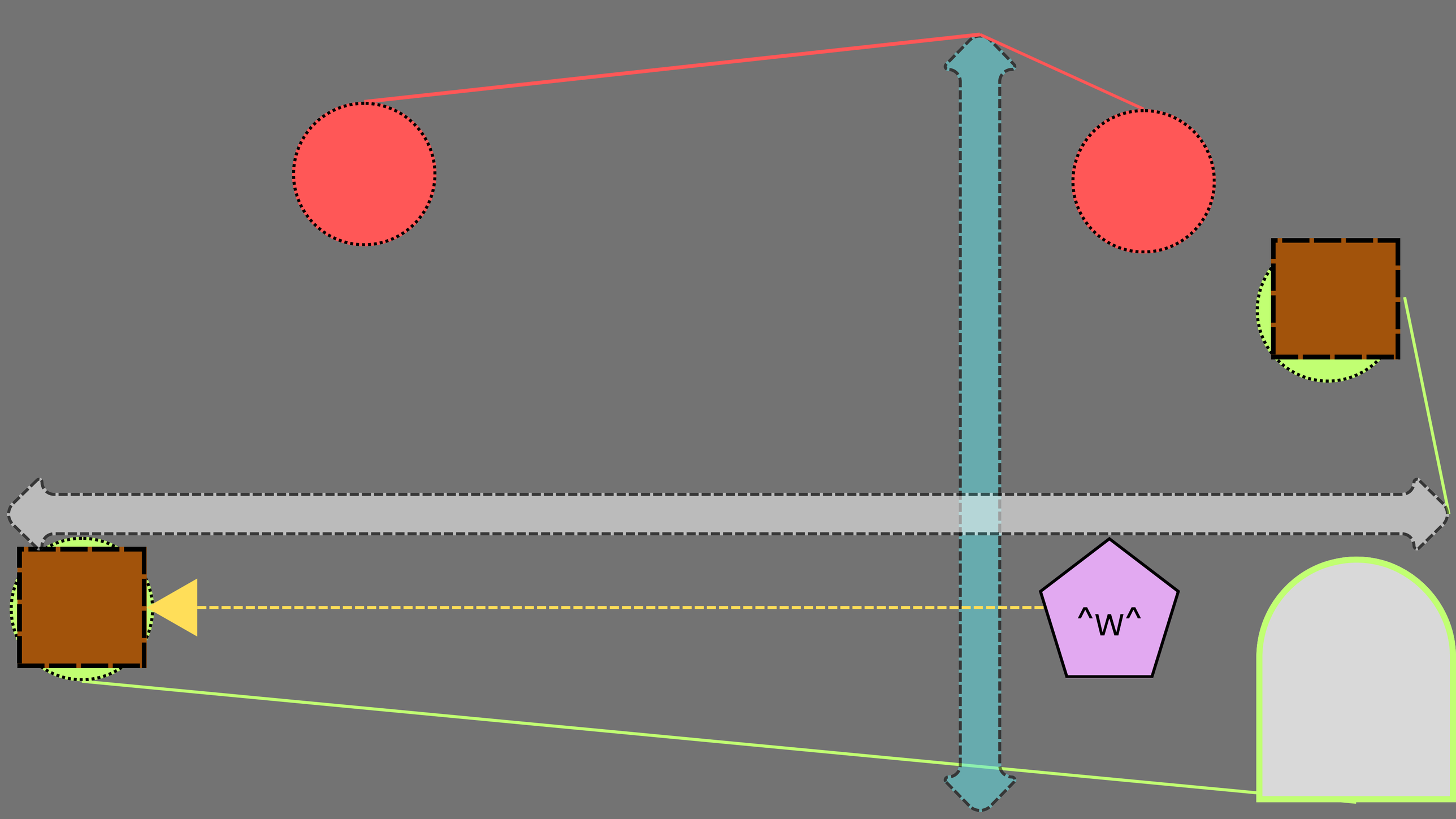
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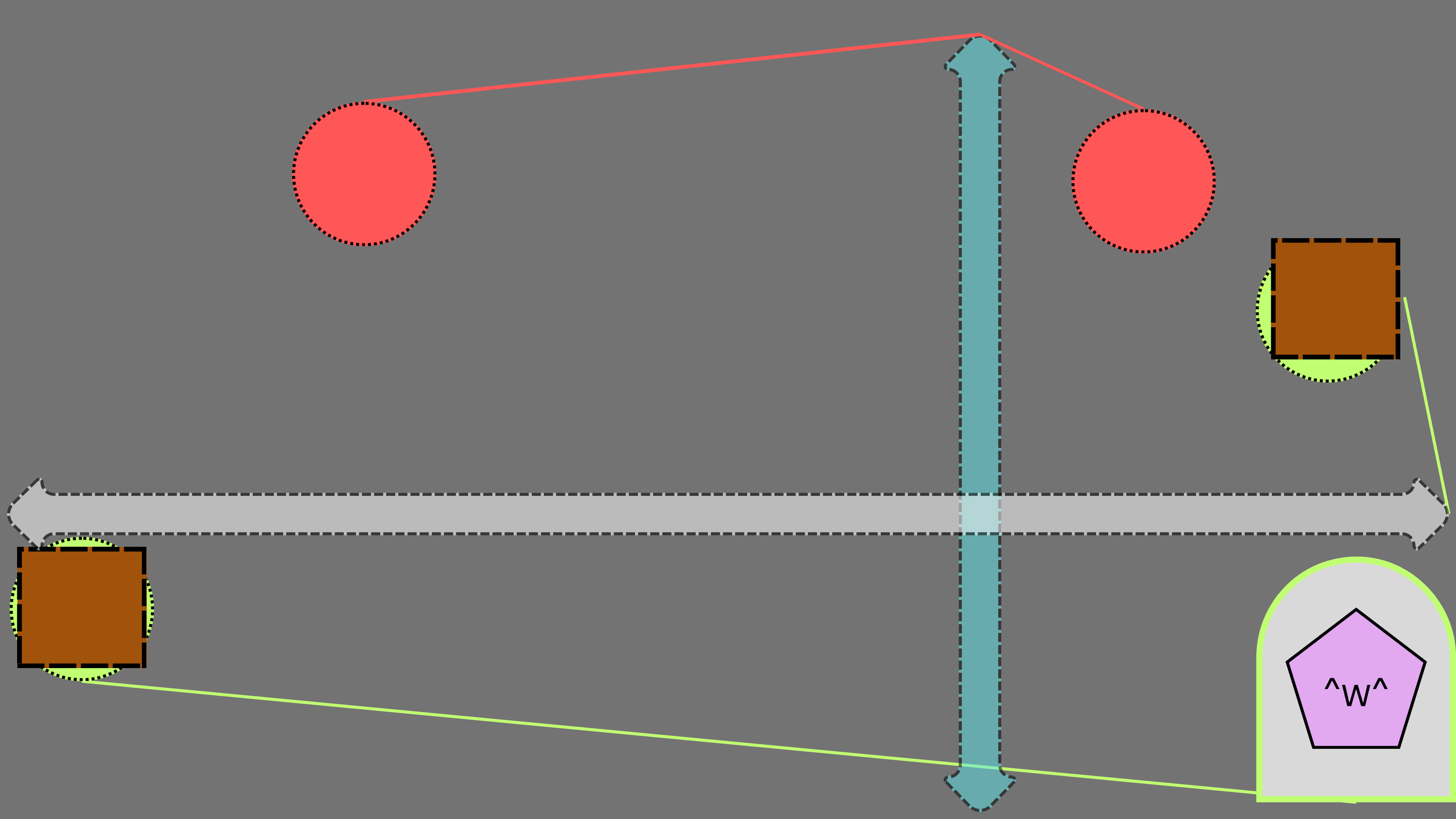
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Assignment 2: Systemic Games with Emergent Systems

Course: GAME 10017

Game Systems Design Professor: Raphaël Tétreault

Designer: Alice Bellefeuille

Date: 2026/02/21

Revisions Table

Version	Date	Author	Revisions
1.0	2026-02-21	Alice Bellefeuille	Initial version

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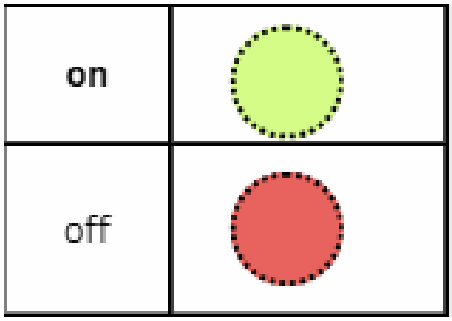
Q1. Justify Objects, Properties, Behaviours, and Relationships

For each mechanic, justify why you designed these specific objects, properties, behaviours, and relationships.

1.0 Assumptions

I did not list every game mechanic, the player moving, moving the box, shooting an arrow, and arrow moving the box are not included in this doc.

1.1 Pressure Button



For the Pressure Button, its purpose is to activate / deactivate other objects, It is very simple but is what is used to alter other objects. The player can see what the button is connected to via the wire.

When its state is changed it will check if there are any connected wires and if so, change their state to match its own.

1.2 Lightning gate

	Light blue	dark blue
on		
off		

This is used to block the player from moving through it and to stop boxes from going through it, arrows can still move through it which allows for some interesting gameplay. The 2 colours are to differentiate between each one as there are only 2. When the lightning gate is off it acts as if it isn't there at all and the player can walk through it and move boxes through it.

When a connected wire sends a signal to this that its signal has changed, it will check all connected wires and if any of them are activated it will be off, otherwise it will be on. In this way it acts as an or gate

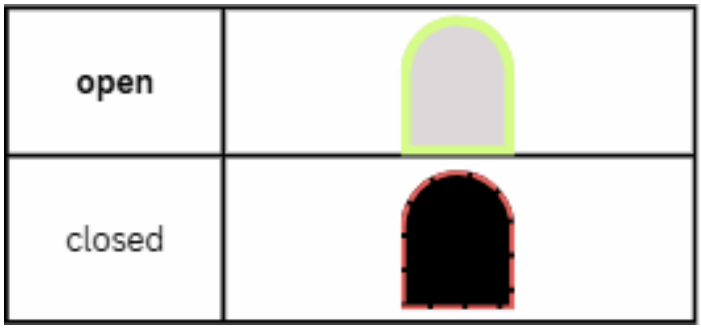
1.3 Wire

on	
off	

This is used to connect objects physically and mechanically. The player can’t directly interact with it, it's more of a visual aid.

It will be linked to a button and an object and when its state is changed it will have the object check its connected wires and adjust its state based on that.

1.4 Exit Door

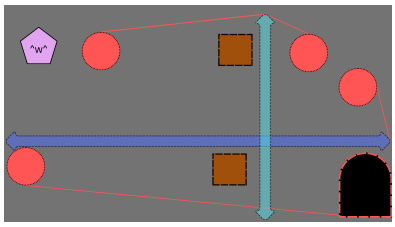
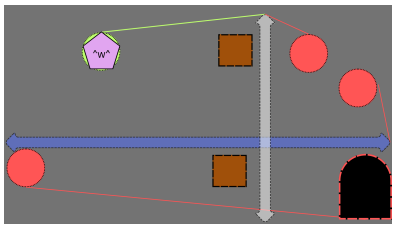
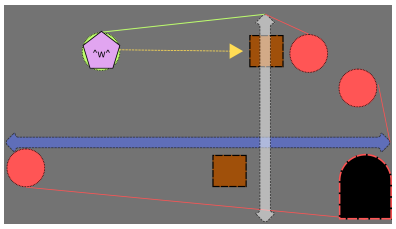
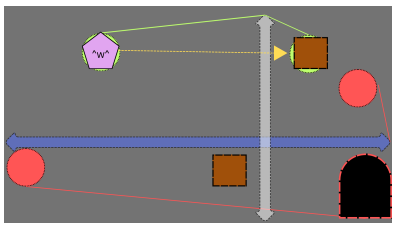
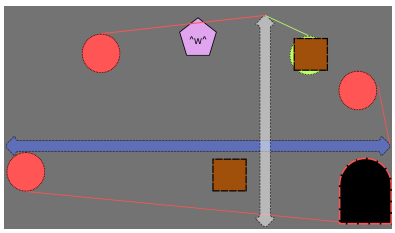
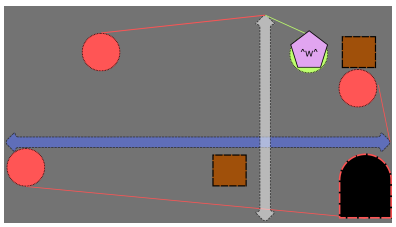
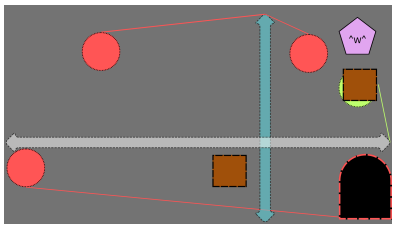
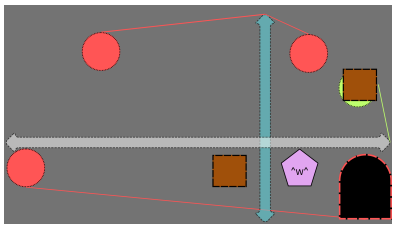
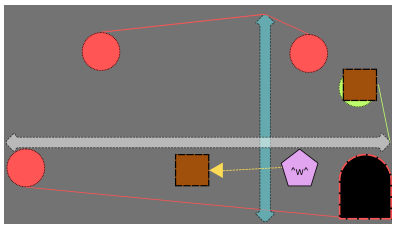


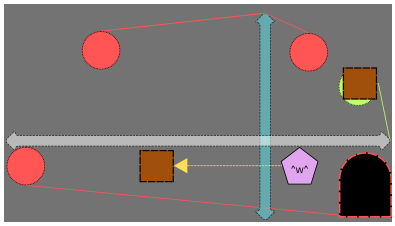
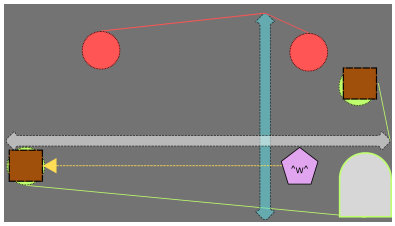
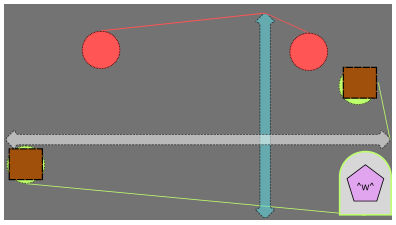
This is the exit door, it's how the player enters the next level, I don’t have too much to say about it, its colour and outline indicate which state its in.

Similarly to the lightning gate its state is changed based on connected wires, it however needs all connected wires to be activated in order to open, in this way it functions as an and gate.

Q2. Justify Emergence, Gameplay Possibility Space

Overall, justify how you used a minimal number of objects with a sufficient number of properties to generate a wide possibility space.

		
Starting scene	The player steps on the button to disable the laser gate	The player shoots an arrow at the box to move it
		
The player shoots until the box is on the button	Player moves right	The player moves the box off the button
		
Player moves the box onto the other button to disable the other gate	Player moves down	Player shoots the second box

		
repeat	Once the box is on the button the player can leave	End

There is only one solution, but there are places to mess up, but the level will reset if the player dies or destroys the box. Unfortunately I'm not sure how much i actually accounted for a wide possibility space, I did use a small amount of objects though.

There will definitely be some things I would improve and change if I did a second draft.